

Facade material detection

Did the building get the material the design decided on?

Problem. Value-engineering decisions fix facade materials during design. After construction starts, nobody checks systematically that the installed material is the one decided, so substitutions slip through.

Prototype. A YOLOv8s detector finds 6 facade materials in building photos. Its output is compared with the specification and mismatches go to a person for review.

Decision supported. Flag a facade for review; never certify compliance.

Final model (Run 04)

0.394 mAP@50 (val)

0.708 Precision

0.345 Recall

DATASET

Data: 253 open-licence photos, labels cleaned in code

6 classes: brick, cladding_panel, exposed_concrete, glass, painted_render, stone_cladding. One box per continuous material area.

Sources: Wikimedia Commons only (CC0, PD, CC BY, CC BY-SA), dataset released as CC BY-SA 4.0, per-image attribution in **data/attribution.csv**.

Problem found: SAM polygons mixed with boxes (181 files), 3,500 fragments, glass labelled per pane (median 101 boxes per image), stacked classes.

Fix (notebook 01b, deterministic): polygons → boxes, drop fragments, merge same-class areas, resolve stacked labels, drop boxes < 1% of the image, drop 24 images with no label of their own class.

Release v1.1: 322 · 45 · 23 images, **13,730 → 968** training boxes, SHA256 bdcf62ff...4187164.



Before / after cleaning (notebook 01b)

Method: transfer learning, fully reproducible in Colab

Item	Setting
Model	YOLOv8s pretrained on COCO, fine-tuned
Training	50 epochs · batch 16 · 640 px · seed 0 · deterministic
Library	ultralytics 8.4.163 (pinned) · torch 2.11 · Python 3.13
Hardware	Google Colab, Tesla T4 · training 5.7 min
Data	GitHub Release v1.1, keyless download, SHA256 checked
Baseline	COCO model: bicycle, car, bird, nothing — no materials

Run it: README badge → T4 GPU → Disconnect and delete runtime → Run all. No accounts or API keys.

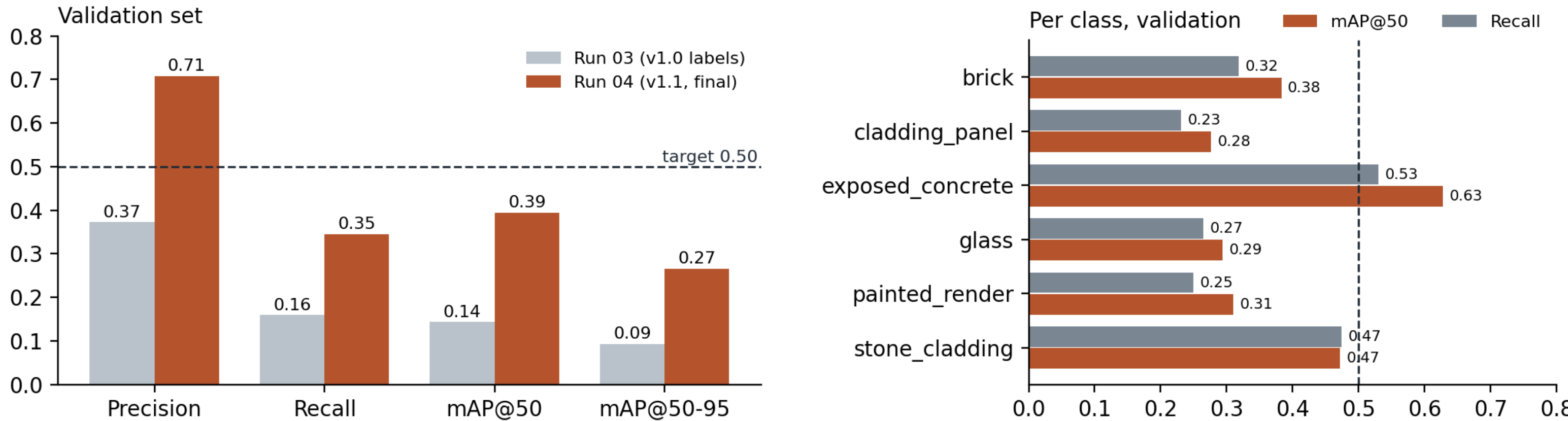
Exploration before the final model: RF-DETR on Roboflow (mAP@50 0.30, ~50 images); YOLO26-S failed (0.001) on corrupted mixed labels; Run 03 YOLOv8s on uncleaned labels 0.14.



Baseline: COCO model sees objects, not materials

RESULTS

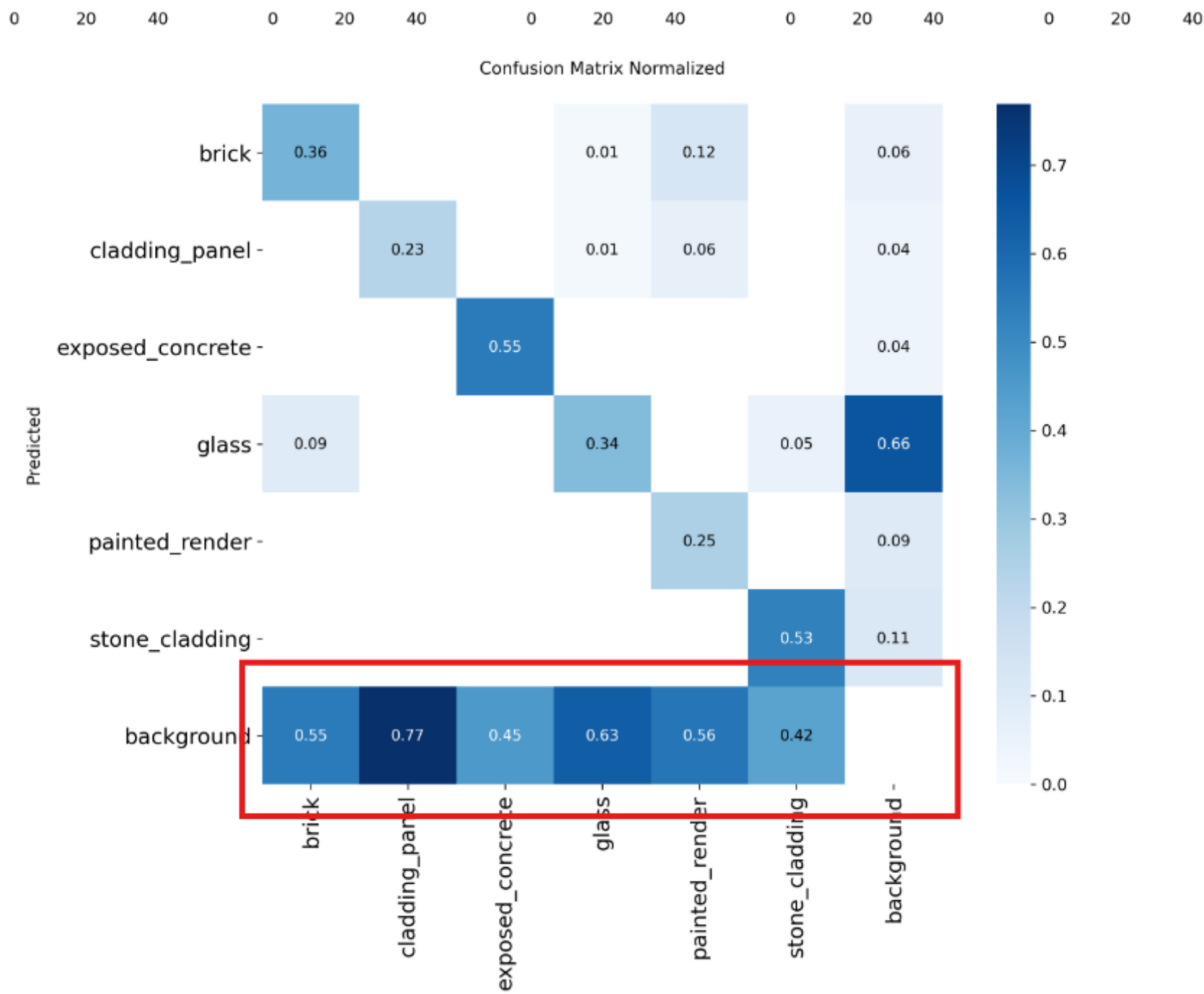
Results: precision 0.708, recall 0.345: recall below target



	Precision	Recall	mAP@50	mAP@50-95
Validation (45 img)	0.708	0.345	0.394	0.265
Test (23 img)	0.480	0.319	0.315	0.248
Target	≥ 0.50 ✓	≥ 0.50 ✗	≥ 0.50 ✗	—

Cleaning the labels almost tripled mAP@50 with the same model. **exposed_concrete** is usable (0.63 val, 0.78 test); **cladding_panel** and **glass** are weakest.

Evidence: validation and new-image predictions



Normalised confusion matrix, validation (columns = true class)

Read the bottom row: the share of each material predicted as background, i.e. missed. It is the largest cell for every class except exposed_concrete and stone_cladding.

Read the right column: detections where nothing was labelled. 66% are glass, mostly real windows in partly labelled images.

Prediction images: results/yolov8s_clean/evidence/ in the repository.

Error analysis: the model misses more than it invents

	Case	Why
FP1	glass fired on unlabelled areas (66% of background detections).	mostly real windows missing from partly labelled images (label gap A6).
FP2	stone_cladding on roof edges and sills (11% of background detections).	some stone labels outline roofs (A4).
FP3	painted_render called brick in 12% of cases.	warm-coloured render with joints; few render examples after cleaning.
FN1	cladding_panel: 77% missed.	fewest examples (13 validation boxes) and very varied metal finishes that look like glass or sky.
FN2	glass: 63% missed.	two object scales after merging (curtain wall vs punched windows) and sky reflections.
FN3	brick on the test split: recall 0.06.	frame-filling walls under-represented (E1); only 7 test images.

3 next data fixes, by impact: (1) relabel to one standard, starting with the 24 removed images and partly labelled ones; (2) at least 80 images per class, most for cladding_panel and painted_render; (3) hard cases on purpose: glare, shadow, occlusion, look-alike materials.

Governance and next steps

Screening aid, not certification. Flags go to an architect or facade consultant; the model never approves payment or handover and is never used for fire or life-safety judgements.

False negatives cost more than false positives: a missed material is a substitution nobody reviews. In use, lower the confidence threshold to trade precision for recall.

Privacy: public, openly licensed photos only; no client or site images. Faces and plates to be blurred in the next dataset version.

Licences: code MIT · dataset CC BY-SA 4.0 · weights AGPL-3.0 (Ultralytics).

Next steps

1. Relabel in Roboflow to the class rules and publish v1.2.
2. Collect 30+ more cladding_panel and painted_render facades.
3. Retrain; pick the confidence threshold on validation for recall ≥ 0.75 on glass, cladding_panel, stone_cladding, exposed_concrete.
4. Link detections to BIM material parameters (IFC) for the FMP value-engineering loop.